Worksheet 3

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1. You've applied to be an operations manager for Coca-Cola Consolidated. In the interview, they ask you we have two different systems of filling a can or a bottle. Which process is working better (in the sense of consistency)?

   i. System 1: A sample of 50 12 oz bottles of Coke which are filled to a number between $11 - 13$ oz. The average of this sample is 12 and the standard deviation is 0.33.

   ii. System 2: A sample of 120 8 oz cans of Coke filled to a number between $7.3 - 8.7$ oz. The average of the sample is 8 and the standard deviation is 0.23.

$$CV_I = \frac{sd_I}{\bar{x}_I} = \frac{0.33}{12} = 0.0275 \Rightarrow CV_{Syst\ I} < CV_{Syst\ II}$$

$$CV_{II} = \frac{sd_{II}}{\bar{x}_{II}} = \frac{0.23}{8} = 0.02875 \Rightarrow Sys\ I \text{ better.}$$

2. A random sample of 10 students were asked how far they live from UMBC. These were their answers, rounded to the nearest half-mile:

$$1, 2.5, 1.5, 2, 0, 5, 3, 3.5, 1, 0.5 \qquad n = 10$$

We calculated the sample mean (2) in the previous worksheet.

   a. Compute the sample standard deviation.

   b. Compute the coefficient of variation.

$$sd = \sqrt{Var}$$

$$Var = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

a

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
1	-1	1
2.5	0.5	0.25
1.5	-0.5	0.25
2	0	0
0	-2	4
5	3	9
3	1	1
3.5	1.5	2.25
1	-1	1
0.5	-1.5	2.25

$$\sum_{i=1}^{10} (x_i - \bar{x})^2 = 21$$

$$Var = \frac{1}{10-1} \cdot 21 = \frac{21}{9} = \frac{7}{3}$$

$$sd = \sqrt{Var} = \sqrt{7/3} = 1.53$$

$\boxed{b}$ $CV = \frac{sd}{\bar{x}} * 100 = \frac{1.53}{2} * 100 = 76.5\%$

$\boxed{N}$

3. A small sample of height (in cm) is collected as follow:

$$164, 148, 137, 157, 173, 156, 177, 172.$$

$n = 8$

a. Determine the location and values of the first, second and third quartiles.

b. Calculate the IQR.

c. Determine the location and value of the 60th percentile. What does the value signify?

Q_1 Q_2 Q_3
P_{25} P_{50} P_{75}

(a) First step : Order : 137 148 156 157 164 172 173 177

$Loc_p = (n+1) * \frac{P}{100} = 9 * \frac{25}{100} = 2.25$

$\boxed{Q1}$

obs 2 obs 3 (ordered) obs 2

$148 + \underbrace{(156 - 148) * 0.25}_{2} = 150 = Q_1 = P_{25}$

$\boxed{Q_2}$ $Loc_{50} = (n+1) * \frac{50}{100} = 9 * 0.5 = 4.5$

$157 + (164 - 157) * 0.5 = 157 + 3.5 = 160.5$

$\boxed{Q_3}$ $Loc_{75} = (n+1) * \frac{75}{100} = 9 * 0.75 = 6.75$

$Q_3 = 172 + (0.75 * (173 - 172)) = 172 + 0.75 = 172.75$

$\boxed{b}$ $IQR = Q_3 - Q_2 = 172.75 - 150 = 22.75$

$\boxed{C}$ $P = 60$

$Loc_{60} = (n+1) * \frac{60}{100} = 9 * 0.6 = 5.4$

$164 + 0.4(172 - 164) = 164 + 3.2 = 167.2$

$= P_{60}$

60% of data less than equal to 167.2

40% of data greater than 167.2

4. The following data has mean income and housing for 10 cities in Florida. Values are in dollars ($) and rounded to the nearest thousand.

City	Income ($\boldsymbol{x}$)	Housing (y)
A	26	109
B	29	97
C	25	115
D	28	99
E	38	122
F	32	145
G	25	100
H	22	76
I	29	113
J	42	144

a. Calculate the correlation coefficient between x and y. What can you conclude about the relationship between the 2 variables?

b. Calculate the least square line.

c. Calculate the coefficient of variation

5. A sample of 30 observations has a standard deviation of 4. Find the sum of squared deviations from the sample mean.